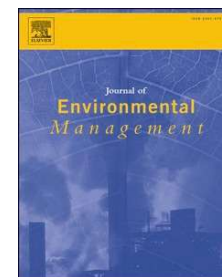


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Research article

Enhancing harmful algal bloom predictions through integrated modeling of turbidity and nutrient dynamics in monsoon climate reservoirs

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ABSTRACT

Effective management of harmful algal blooms (HABs) in dam reservoirs is crucial for water resource sustainability, particularly in monsoon climates, such as South Korea, where approximately 55 % of the domestic water supply depends on these reservoirs. To address the critical need for accurate HABs prediction, an innovative three-dimensional hydrodynamic and water quality model that integrates turbidity with traditional nutrient-based approaches was developed. This study focused on the Daecheong Reservoir in the Geum River Basin, simulating the density currents of turbid water during rainfall events, and quantifying their impact on algal (Chl-a) concentrations. The results revealed that turbidity-induced light limitation significantly reduced Chl-a concentrations by 44–58 % across the reservoir, with the effects intensifying from the riverine and transitioning to lacustrine zones. The model improved light extinction coefficient estimations by incorporating multiple water quality parameters, including suspended solids, and outperformed the conventional single-parameter Secchi depth approach. The integrated modeling approach improved the prediction accuracy by 51–70 % (RMSE-ERR, Root Mean Square Error-Error Reduction Rate) and 52–67 % (MAPE-ERR, Mean Absolute Percentage Error-Error Reduction Rate) when compared to conventional nutrient-only models, while also quantifying seasonal variations in turbidity-nutrient interactions during monsoon events. These findings demonstrate that traditional nutrient-focused HABs modeling and management strategies are insufficient, as they overlook the critical role of turbidity in algal growth in monsoon-affected reservoirs. This novel modeling approach advocates comprehensive HABs management that considers turbidity-nutrient interactions, thereby contributing to more effective and sustainable water quality practices in monsoon-affected reservoirs under changing climatic conditions.

1. Introduction

Water resource management presents unique challenges in regions characterized by seasonal precipitation patterns. This challenge is particularly pronounced in South Korea, which is situated in a temperate monsoon climate zone. With approximately 55 % of its water supply dependent on large reservoirs such as multipurpose dams, the nation faces a critical need for effective reservoir water quality management. Consequently, water quality modeling techniques play a crucial role as essential support tools for water management. The importance of this issue is underscored by the prevailing trophic states of several multipurpose dam reservoirs in South Korea, which range from mesotrophic to eutrophic. These conditions are often associated with various water quality issues and are primarily driven by excessive algal growth (Kim et al., 2001; Anderson et al., 2002; Ahn et al., 2015; Watson et al., 2016). Although algae are essential primary producers in aquatic ecosystems,

their overgrowth can lead to the severe degradation of water quality. Notable examples include harmful algal blooms (HABs) and hypoxic conditions (Franchini et al., 2017; Pal et al., 2020; Zhao et al., 2020).

Algal growth in water bodies is influenced by various limiting factors, including nutrients such as phosphorus and nitrogen, water temperature, and light (Hansson, 1988; Wetzel, 2001). These limiting factors act as bottom-up controls, whereas interspecific competition and predation at higher trophic levels serve as top-down controls (Huang and Qi, 1997; Rudman et al., 2015). Light is an essential energy source for photosynthesis. Light varies with climate and weather conditions as well as with water clarity, typically measured by Secchi depth, and plays a crucial role in algal growth (Cole et al., 1992; Scheffer, 1998; Litchman et al., 2003). In reservoirs within monsoon climate regions, suspended solids (SS) are the primary factor inhibiting algal photosynthesis during summer. Suspended solids (SS), which enter reservoirs from watersheds with rainfall, cause high turbidity and reduce water transparency,

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thereby hindering algal growth (Brown, 1984; Lind et al., 1992). In particular, highly turbid water influxes associated with extreme rainfall events can persist in dam reservoirs for extended periods, diminishing light penetration, and impeding photosynthesis (Brown, 1984; Sobolev et al., 2009). Owing to the considerable spatiotemporal variability of these SS, a comprehensive analysis of the entire water body, rather than point-based experimental studies, is necessary.

Reservoirs exhibit spatially diverse physical and chemical characteristics depending on their size, shape, and watershed properties and are substantially influenced by artificial operational environments. In artificial dam reservoirs, gradients in water temperature, light intensity, and nutrient concentrations occur across the riverine, transitional, and lacustrine zones (Kimmel and Groeger, 1984). Consequently, multidimensional numerical modeling, which accounts for this complexity, is highly efficient for analysis (Jeznach et al., 2017; Sadeghian et al., 2018). Numerical modeling techniques for multidimensional analysis of hydrodynamics, water quality, and aquatic ecosystems are continuously being improved. Major models include EFDC (Hamrick, 1992), CE-QUAL-W2 (Cole and Wells, 2006), Delft3D (Lesser et al., 2004), and AEM3D (Hodges and Dallimore, 2019). These models are actively used in policy implementation, decision making, impact assessment, prediction, and evaluation (Nguyen et al., 2017; Tranmer et al., 2020).

In water quality models, algal prediction is performed by considering physiological parameters such as algal growth, mortality, and respiration, as well as limiting factors such as light, water temperature, and nutrients (Whitehead et al., 2015; Martin et al., 2021). However, many previous studies have mainly focused on water temperature and phosphorus, whereas light limitation has been studied using relatively simple models, such as one-dimensional or logistic statistical models (Scheffer et al., 1993; Gelman and Hill, 2007; Scheffer, 2009). May et al. (2003) studied the decrease in algae caused by increased SS, whereas Chen et al. (2015) investigated the effects of temperature and light on cyanobacterial growth. Watson et al. (2016) revealed that light and temperature are crucial for initial growth and Andersen et al. (2022) simulated changes in algal biomass according to turbidity. Despite the well-documented importance of light penetration in algal growth dynamics, previous water quality modeling studies on monsoon-affected reservoirs have predominantly focused on nutrient-driven eutrophication, particularly phosphorus loading, during rainfall events. These models have largely overlooked the critical role of SS in modulating light availability, which is a significant oversight given the characteristic intense rainfall and subsequent SS influx in monsoon regions. Traditional approaches typically rely on parameter adjustment (e.g., growth rates, mortality, and respiration) to improve model performance, rather than explicitly incorporating spatiotemporal variations in light limitation caused by SS transport and distribution. To address these limitations, this study introduces a novel modeling framework that integrates SS-induced light attenuation with conventional nutrient dynamics with the aim of enhancing HABs prediction accuracy in monsoon-affected reservoirs by capturing the complex interactions between turbidity and algal growth during rainfall events.

Therefore, this study aimed to simulate the density current behavior of turbid water during rainfall events in the Daecheong Reservoir, located in a monsoon climate zone, using a three-dimensional (3D) numerical model to analyze the spatiotemporal variability of algal growth-limiting factors (light, water temperature, and nutrients) within the reservoir and to quantitatively assess the impact of light limitation caused by turbid water on algal prediction. In addition, this study proposes a modeling method that can simultaneously consider turbid water and algal growth by overcoming the limitations of existing models. Daecheong Reservoir is a critical water source for approximately 3.5 million people. However, the reservoir faces significant challenges due to the persistent turbidity caused by heavy rainfall and HABs, which are exacerbated by eutrophication and long retention times. These issues highlight the importance of effective water quality management for this key resource.

This study is expected to advance water quality modeling techniques by developing a 3D modeling approach that considers the interactions between turbid water and algal growth. This will enable more accurate and realistic water quality predictions and contribute to improved water quality management in dam reservoirs in monsoon climate regions. More effective water quality management strategies could be established by quantitatively understanding the impact of turbid water during rainfall on algal growth. Moreover, by considering light limitation factors in detail, more accurate predictions of algal blooms are possible compared with existing models. Ultimately, the results of this study are expected to provide scientific evidence for the establishment of water quality management policies for dam reservoirs located in monsoon climate regions, thereby contributing to the sustainable use of water resources.

2. Materials and methods

2.1. Description of study site and field monitoring

Daecheong Dam, located approximately 150 km from the Geum River Estuary, was constructed in 1981 for water supply, hydroelectric power generation, and flood control purposes (Fig. 1). The completion of Daecheong Dam resulted in the formation of Daecheong Reservoir, with a total storage capacity of $1490 \times 10^6 \text{ m}^3$ and an effective storage capacity of $790 \times 10^6 \text{ m}^3$. The Daecheong Reservoir receives direct inflows from the main stream of the Geum River and its tributaries, including the Hoein Stream, Seohwa Stream, Juwon Stream, and Pungok Stream. The total catchment area is 3204 km^2 , accounting for 32.4 % of the entire Geum River Basin.

Six water quality monitoring stations were located within the reservoir and provided regular monthly measurements ($\text{PO}_4\text{-P}$ (Phosphate Phosphorus), $\text{NO}_3\text{-N}$ (Nitrate Nitrogen), $\text{NH}_4\text{-N}$ (Ammonium Nitrogen), chlorophyll *a* (Chl-*a*), Dissolved Oxygen (DO), Total Organic Carbon (TOC), Total Nitrogen (T-N), Total Phosphorus (T-P), and SS) as public information. Meteorological data were obtained from the Automatic Weather System located within the reservoir watershed, which offers hourly and daily measurements of atmospheric pressure, temperature, humidity, wind direction, and wind speed. Additionally, the automated synoptic observation system provided solar radiation and total cloud cover data. A thermistor chain (HOBO Water Temp Pro, ONSET, Bourne, MA, USA) was installed in front of the dam to continuously measure the water temperature at various depths from the surface to the bottom (38 m). Hourly automated monitoring data (water temperature and turbidity) were collected from a corporation self-operated station located at the reservoir inlet at the Korea Water Resources Corporation (K-water). The initial water level and dam discharge data were obtained from the K-water daily dam operation hydrological records (Fig. 1, Table S2).

Daecheong Reservoir can be broadly divided into three zones: riverine, located in the upstream area of the reservoir, furthest from the dam with water depths of 4–8 m; transition, situated in the middle reaches of the reservoir with water depths of 10–20 m; and lacustrine, located in the downstream area adjacent to the dam with an average water depth exceeding 30 m. In this study, representative points were selected based on the locations of national water quality monitoring stations. Daejeongli (DC6) and Hoenam (DC5) were selected to represent the riverine and transitional zones, respectively. The dam front (DC2) was selected to represent the lacustrine zone. These selections considered the distinct environmental characteristics of each spatial zone.

2.2. HABs occurrence and characteristics in the study area

The Ministry of Environment in South Korea has been operating a HABs alert system since 1997 to monitor and manage algal blooms in reservoirs, rivers, and recreational water areas. This system ensures a

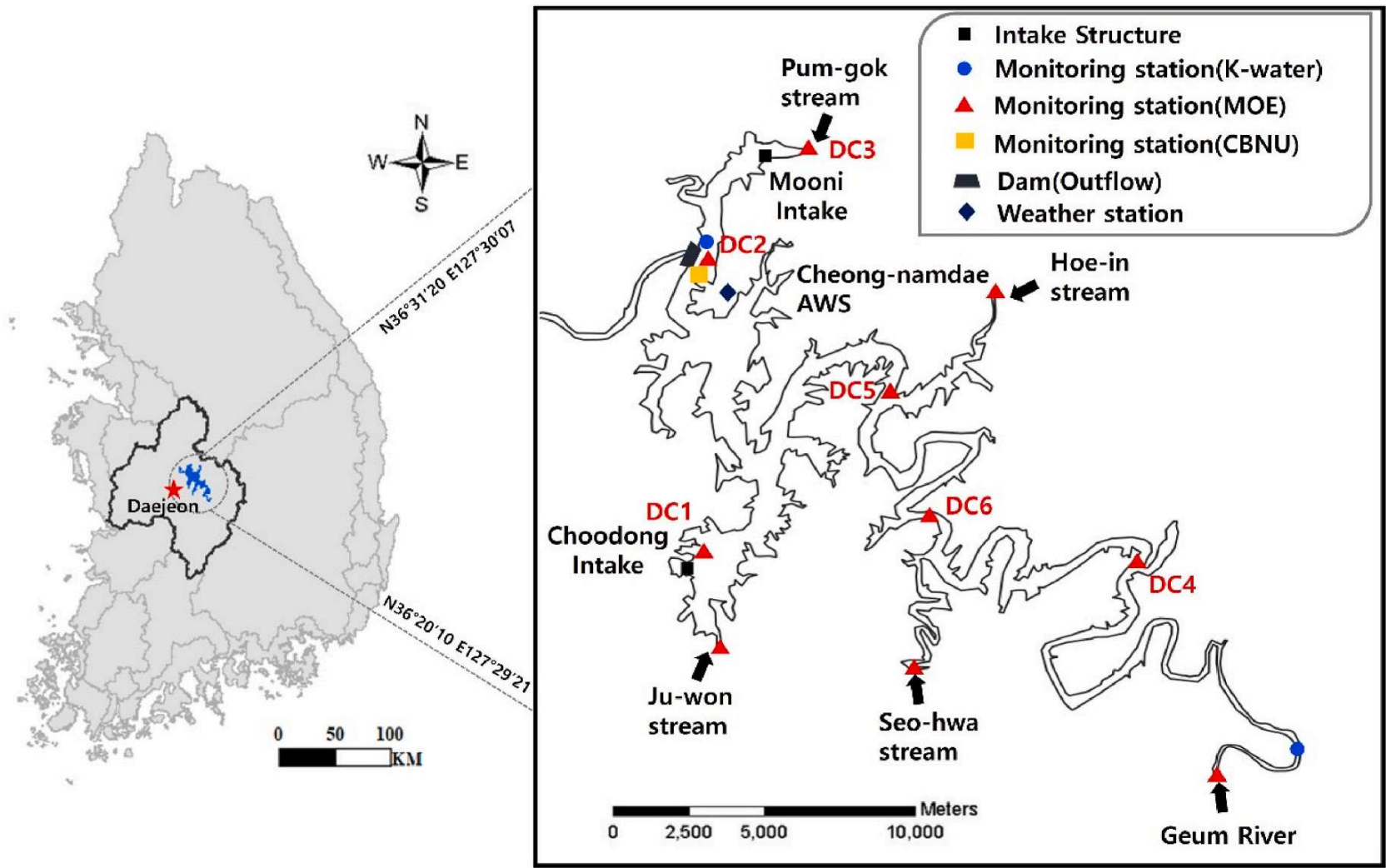


Fig. 1. Study site and monitoring station locations in the Daecheong Reservoir, South Korea.

stable supply of drinking water and protects the public from harmful algal toxins during recreational activities. Regular monitoring was conducted, and if the algal levels surpassed the established thresholds, an alert was issued prompting necessary preventive measures. Specifically, an alert was triggered when harmful cyanobacterial cell counts exceeded the standard limit for two consecutive weekly measurements (Ministry of Environment Korea, 2019). Harmful cyanobacterial cell counts refer to the sum of four genera: *Microcystis*, *Dolichospermum* (*Anabaena*), *Aphanizomenon*, and *Oscillatoria*. The criteria for issuing and lifting the alerts are listed in Table S1.

Daecheong Dam plays a crucial role in providing drinking water to approximately 3.5 million people. Accordingly, an HABs alert system was operated at three locations within the reservoir: DC1, DC3, and DC5. HABs alerts were issued for the Daecheong Reservoir annually, except in 1999 and 2014. The long, meandering shape of the reservoir makes it structurally vulnerable to algal blooms, which are primarily caused by the prolonged retention of pollutants introduced by heavy summer rainfall (National Institute of Environmental Research [NIER], 2009; Chung and Lee, 2011). In 2018, the HABs alert in Daecheong Reservoir was maintained for an extended period (Table 1). At DC5 site, the maximum harmful cyanobacterial cell count reached 47,190 cells/mL, resulting in the issuance of a warning alert. During this period, the harmful cyanobacteria were predominantly *Microcystis*, with *Dolichospermum* appearing as the secondary dominant species (Ministry of Environment Korea, 2019).

2.3. Three-dimensional (3D) hydrodynamic and water quality model

2.3.1. Aquatic ecosystem model

The Aquatic Ecosystem Model (AEM3D) is a 3D hydrodynamic-water quality model developed by HydroNumerics (www.hydronumerics.com.au) based on the Estuary, Lake and Coastal Ocean Model (ELCOM) and Computational Aquatic Ecosystem Dynamics Model (CAEDYM), which were originally developed by the Centre for Water Research at the University of Western Australia (Hodges and Dallimore, 2019; Hipsey et al., 2006). The model is a coupled hydrodynamic and water quality

Table 1

Summary of the HAB alert records for the Daecheong Reservoir monitoring stations (DC1, DC3, DC5) in 2018, including cyanobacterial cell counts, alert levels, and dominant genera.

Station	Number of alert days (day) ^a		First occurrence of cell counts exceeding 1000 cells/mL and the time of appearance	Maximum cell count (cell/mL) and occurrence date
	Caution	Warning		
DC1	63 (First issued on 8/22)	0	7716 (8/13)	8574 (9/17)
DC3	77 (First issued on 8/8)	0	5874 (7/30)	8036 (8/6)
DC5	60 (First issued on 8/16) (Downgraded on 9/22)	9 (First issued on 9/12)	4600 (8/6)	47,190 (9/3)

^a Alerts were triggered based on the South Korea's HABs alert system criteria (Table S1), which monitor harmful cyanobacteria exceeding threshold levels for consecutive weeks.

model that allows for integrated analysis of hydrodynamics, water quality, and ecological processes. The hydrodynamic model includes momentum, continuity, free surface, and advection-diffusion equations for substances. The AEM3D model analyzes key state variables, including inflow nutrient loading, outflow losses, convection, mixing driven by hydrodynamic forces, and internal production processes. The specific variables examined in this study included the concentrations and cycling of carbon, nitrogen, phosphorus, silicon, dissolved oxygen, and size-fractionated suspended solids as well as the dynamics of various algae species (Fig. S3).

2.3.2. Light extinction coefficient

The light extinction coefficient represents the degree to which a substance scatters and absorbs light, and it varies depending on factors such as suspended solids, algal biomass, and dissolved organic matter. This can be derived using the relationships among light attenuation, SS,

turbidity measurements, or Secchi disk depth (Williams et al., 1980; Brown, 1984; Chapra, 1997; Armengol et al., 2003; Hipsey et al., 2006; Caffrey et al., 2007). In reservoirs, water transparency can be influenced by various factors including SS, algae, aquatic plants, and organic matter. Solar radiation undergoes several processes when it reaches the water surface. A portion was reflected, and the remainder was absorbed. During this process, particles within the water body scatter and reflect light, thereby reducing the amount of light absorbed (Kirk, 1994).

In the AEM3D model, photosynthetically active radiation (PAR) is calculated spatiotemporally using the Beer-Lambert law (Swinehart, 1962). The model incorporates the light extinction coefficient, which accounts for the variability of phytoplankton, inorganic and detrital particles, and dissolved organic carbon. This calculation is given by Equation (1):

$$\gamma_{PAR} = \gamma_w + \sum_a^{N_A} K_e^{A_a} A_a + \sum_d^{N_D} K_e^{DOC_d} DOC_d + \sum_d^{N_D} K_e^{POC_d} POC_d + \sum_s^{N_S} K_e^{SS_s} SS_s \quad (1)$$

Here, γ_w represents the background value for clear water, while K_e denotes the rate at which the extinction coefficient increases with the concentration of variables. Different K_e values were applied for DOC, POC, algae, and SS according to size, as shown in Table S5 in the Supplementary Data. This incorporates the spectral absorption and scattering coefficients of various water constituents at different wavelengths. This approach effectively captures the spatiotemporal variability influenced by environmental factors such as turbid inflows during rainfall events and summer algal blooms.

2.3.3. Model validity assessment

To evaluate the suitability of the simulation results, we employed statistical indicators to measure the agreement between the observed and simulated values of the Index of Agreement (IOA), Root Mean Square Error (RMSE), and Mean Absolute Percentage Error (MAPE) (Table S4). The IOA, a measure of predictive accuracy, ranges from 0 to 1. Values closer to 1 indicate that the model predictions better reflect the actual phenomena, whereas those approaching 0 indicate a lower concordance between observations and predictions (Willmott, 1981). RMSE quantifies the average magnitude of the prediction errors by calculating the square root of the mean squared differences between the simulated and observed values. Lower RMSE values indicate higher reliability of the simulation results. The MAPE expresses the average absolute percentage difference between the observed and simulated values. MAPE is calculated by dividing the absolute difference between the actual and predicted values by the actual value and then expressing this as a percentage. Similar to the RMSE, MAPE values closer to 0 suggest higher model reliability (Chai and Draxler, 2014).

2.4. Preparation of input data

2.4.1. Initial conditions and boundary conditions

In this study, an AEM3D topographic grid was constructed using the survey results and digital terrain data provided by K-water (Korea Water Resources Corporation, 2006). The numerical grid was divided into 100 m × 100 m cells in the longitudinal (x) and transverse (y) directions, with 1 m intervals in the depth (z) direction. A Z-grid system was employed for the vertical grid, resulting in 187,628 cells.

The model was established using input data from 2018 (Figs. S1 and S2) followed by calibration. The essential input data for model construction included the inflow boundary, meteorological, and initial condition data. Water quality input data were obtained from water quality monitoring stations of the Ministry of Environment at the inflow boundaries and within the reservoir. Meteorological data were sourced from disaster prevention and synoptic weather observation data provided by the Korea Meteorological Administration (Fig. 1; Table S2). For the boundary conditions, the inflow and outflow rates were derived

from the hydrological data of the dam operation. The inflow of the four unmeasured tributary streams was estimated using watershed area ratios. The model simulates carbon, nitrogen, and phosphorus in refractory, labile, dissolved, and particulate forms. Accordingly, the input data for the mainstream and four tributary inflows were derived from the Ministry of Environment's water quality monitoring stations, which provided measurements of TOC, TN, TP, NH₄-N, NO₃-N, and PO₄-P. Specifically, the values corresponding to each organic matter form were calculated by multiplying the TOC by the fraction representative of that form, while organic nitrogen and phosphorus were estimated by subtracting the contributions of dissolved inorganic nitrogen (NH₄-N and NO₃-N) and dissolved inorganic phosphorus (PO₄-P) from TN and TP, respectively. Based on earlier analyses of the target watershed, the dissolved and particulate fractions were determined to be 75 % and 25 % of the total organic matter, respectively, while the labile and refractory components were apportioned as 30 % and 70 %, respectively. This approach aligns with prior studies indicating that, for freshwater streams and reservoirs, dissolved matter typically constitutes 60–80 % of the total, whereas labile matter accounts for 20–40 % (Perdue and Ritchie, 2003). To account for the high concentrations of organic matter and phosphorus during rainfall periods that were not captured by regular monthly measurements, weighting factors were applied during rainfall events: 1.5 TOC and 3.1 T-P. These factors are based on a previous study (Geum River basin management Korea, 2021).

This model simulates HABs by categorizing algae into three distinct groups: diatoms, green algae, and cyanobacteria. Chl-a concentration served as the primary measurement unit in the simulations. However, the water quality monitoring network provides data only for total Chl-a, necessitating a method to partition this total into group-specific Chl-a concentrations. To achieve this, previous studies conducted in the same study area were referenced (Geum River basin management Korea, 2021). Based on these findings, the following Chl-a percentages were applied to each algal group: 43 % for diatoms, 46 % for green algae, and 11 % for cyanobacteria. These proportions were derived from field observation data collected from the same reservoir and incorporated into the model input data construction process. These values were then used to partition the total Chl-a concentration into group-specific values to ensure a realistic representation of the algal community structure. To account for the unique physical, biological, and chemical characteristics of each algal group, we implemented distinct parameters for diatoms, green algae, and cyanobacteria in the simulations (Table S5). This parameterization was based on the work of Chung et al. (2014) and supported by the literature (United States Environmental Protection Agency [USEPA], 1985; Cole and Wells, 2006; National Institute of Environmental Research, 2009; Hipsey et al., 2006; Hodges and Dallimore, 2019), which collectively demonstrate the importance of group-specific modeling in understanding the dynamics of HABs.

2.4.2. Inflow temperature prediction

The inflow water temperature is crucial input data that determines the behavior of turbid-density currents in reservoirs. However, when using data with insufficient measurement frequencies or numerous missing values, accurately analyzing the density currents in reservoirs becomes challenging. This is particularly problematic when capturing rapid temperature decreases during periods of intense summer rainfall. To address this issue, we developed a random forest model (Breiman, 2001), which is a machine learning approach capable of accurately predicting inflow water temperatures using minimal input variables.

After performing preprocessing tasks, such as handling missing values and removing outliers through exploratory data analysis, on the collected data, the random forest model was trained. The model was trained using supervised learning with flow and meteorological data from 2008 to 2017 as independent variables, and water temperature data as the dependent variable. The predictive performance of the model was evaluated using 2018 measured data as the test dataset. To prevent overfitting during model development, a 10-fold cross-validation was

repeated three times, and the model was developed using the random forest package in R (Breiman and Cutler, 2022). Additionally, the Recursive Feature Elimination technique was used to optimize the independent variables by removing less important variables. The final input variables used were the flow rate, air temperature, solar radiation, dew point temperature, and humidity.

The model appropriately reproduced the temporal variations of the measured water temperature in 2018, with prediction performance and error metrics of IOA 0.93, RMSE 1.88 °C, and MAPE 9.61 % (Fig. S2). Previously used multiple regression models for inflow water temperature prediction generally exhibited good performance; however, they were developed based on linear relationships between the input variables. This limits their ability to consider nonlinear relationships and reproduce the temperature drops that occur during sudden increases in watershed runoff due to rainfall (Mackey and Berrie, 1991).

2.4.3. Supplementing inflow suspended solids

For operational management, K-Water independently conducts automated measurements of water temperature and turbidity at the reservoir inflow point and in front of the dam. However, numerous missing values were observed during the winter and summer for various reasons. These include equipment loss and measurement sensor fouling caused by intense rainfall as well as poor equipment performance due to low temperatures in winter. Moreover, because the model used SS (mg/L) as the input data, an intermediate process was necessary to convert the continuous turbidity measurement data to SS. This conversion step is essential for preparing data for the model input. Therefore, in this study, we converted the measured turbidity data to SS concentration using a turbidity-SS relationship equation. For periods with missing data, we interpolated the values using an equation describing the relationship between flow rate and turbidity. For the four tributaries, excluding the mainstream, only monthly SS measurement data were provided without turbidity measurements, resulting in a substantial lack of boundary data. Thus, we produced daily SS time series data by applying the flow rate-SS relationship equations for each tributary based on previous research cases (Table S3; Chung et al., 2009; K-water, 2021). To account for the actual sediment characteristics, based on the particle size distribution data analyzed during the rainfall period, particles were classified into three groups: SS1 (smallest), SS2 (medium), and SS3 (largest), with particle sizes of 4, 10, and 20 μm , respectively (Chung et al., 2009). The settling velocities for each SS group were calculated using Stokes' law. To further account for the influence of the particle size on the light dynamics, the model applied distinct light attenuation coefficients (K_e) to each SS group. SS1 (smallest particles) was assigned a K_e of $0.070 \text{ g}^{-1}\text{m}^3\text{m}^{-1}$, while SS2 and SS3 were assigned coefficients of 0.055 and $0.050 \text{ g}^{-1}\text{m}^3\text{m}^{-1}$, respectively (Table S5). These values reflect the increased light scattering and absorption caused by smaller particles as well as their prolonged suspension in the water column, which amplifies their impact on light penetration and turbidity. Chung et al. (2009) in their study further emphasized that each particle size group exhibited distinct settling characteristics in the reservoir, influencing their spatial and temporal distributions. Larger particles settle more rapidly, forming sediment layers in the riverine zone, whereas smaller particles remain suspended for longer periods, affecting turbidity and downstream light attenuation. Sedimentation dynamics are essential for understanding the interactions between sediment transport and reservoir ecology.

3. Results

3.1. Validation of AEM3D model

3.1.1. Water temperature and turbidity density current

Turbid water entering a water body forms density currents because of the temperature differences between the inflow and ambient water body. Therefore, the accurate prediction of water temperature and high reproducibility of thermal stratification are crucial for precisely

predicting the spatiotemporal distribution of water quality parameters such as SS, nutrients, and algae. In this study, we compared the simulated values with the observed data at a point in front of the dam during the 2018 hydrological event to verify the ability of the model to reproduce depth-specific distributions of water temperature and turbidity. The model adequately reproduced seasonal changes in the thermal stratification structure of the Daecheong Reservoir and demonstrated high predictive performance for most of the period (Fig. 2). For water temperature predictions, the model was considered appropriately calibrated, with RMSE ranging from 0.79 to 1.69 °C, and MAPE from 0.95 to 12.06 %.

Although turbidity was slightly underestimated in some periods, the simulated values accurately reflected the trends in the observed data (Fig. 3). Error analysis showed RMSE ranges from 0.67 to 4.05 NTU, and MAPE ranges from 43.7 to 92.8 %. The spatiotemporal distribution of the turbidity within the reservoir was examined using curtain and sheet views (Figs. S4 and S5). In September, the turbid water entering the reservoir from the upstream headwaters was initially transported along the surface, driven by flow inertia, and reached approximately 28 km from the dam. However, dominated by negative buoyancy owing to density differences, it plunged below the reservoir surface and progressed to the dam. The thermocline was located at an elevation of approximately 40–50 m (EL.) above the mean sea level. With the density current forming above the thermocline, the turbid water layer moved along the density current to the dam above EL. 50 m. During this process, SS3 settled predominantly on the reservoir bed, whereas SS1 remained suspended in the middle layer and eventually reached the upstream face of the dam.

3.1.2. Water quality and algal biomass

We present the calibration results by comparing the model's simulation results with the observed values in the time series for water quality parameters (Chl-a, TOC, T-P, PO4-P, T-N, and SS) at representative points within the reservoir: DC6 (representing the riverine zone), DC5 (representing the transition zone in the middle), and DC2 (representing the lacustrine zone near the dam) (Fig. 4). In addition, we calculated and presented the IOA, RMSE, MAPE, and values for the simulation results.

Despite significant spatiotemporal variations in the observed Chl-a values, the model adequately captured these fluctuations, particularly the excessive algal growth characteristics at the inlet during spring (Fig. 4a). Although overestimation occurs in some periods, limitations in calibrating individual algal group concentrations owing to insufficient observational data result in reduced reproducibility for certain algal species during specific periods (Chung et al., 2014). Error analysis for Chl-a shows an IOA of 0.68–0.83, RMSE of 1.11–2.29 mg/m^3 , and MAPE of 29.9–68.9 %. The model appropriately reproduced the spatiotemporal patterns of TOC, which is an indicator of total organic matter. Error analysis shows an IOA of 0.37–0.45, RMSE of 0.450–0.60 mg/L , and MAPE of 12.5–20.9 %, with lower error rates at the DC2 point near the dam (Fig. 4b).

The simulated values for the phosphorus series generally reflected the spatiotemporal variation characteristics of the observed values (Fig. 4c and d). However, reproducibility was lower in some periods because of missing observed data and insufficient resolution of the input data. For T-P, error analysis indicates an IOA of 0.42–0.64, RMSE of 0.02–0.03 mg/L , and MAPE of 35.80–46.39 %. The nitrogen series exhibited no distinct seasonal variation, which was generally reflected in the model (Fig. 4e). However, a sharp decrease in the simulated values is observed at all points after July. This issue stems from the steep decline in TN from July to September in the input boundary data, which were selected from the most suitable river water quality monitoring network points on the mainstream of the Geum River flowing into the Daecheong Reservoir.

Suspended solids (SS) increased temporarily during rainfall events. The model slightly overestimated the SS at the dam inlet, but generally

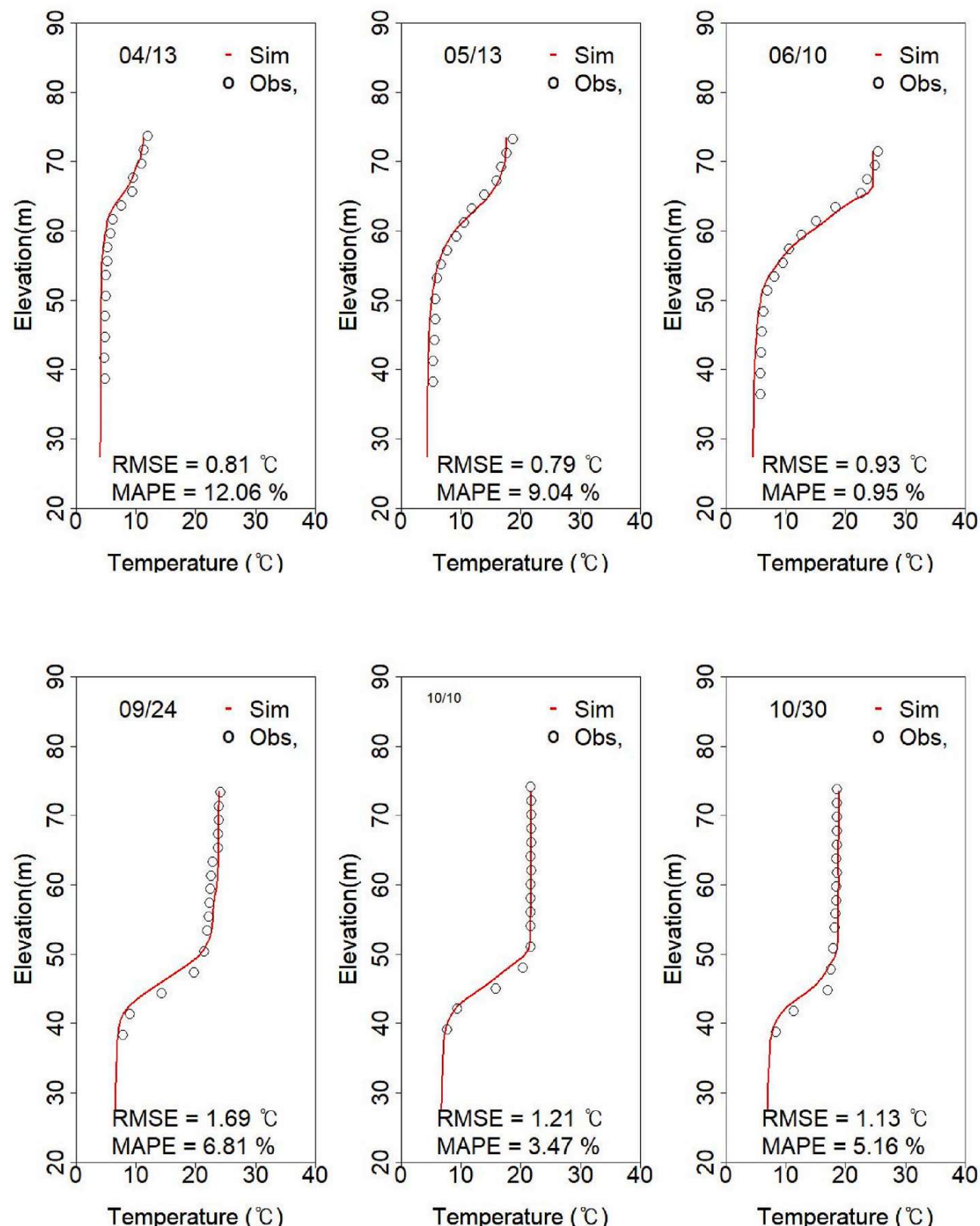


Fig. 2. Comparison of the observed and simulated water temperature profiles at the DC2 station.

reflected the trend of the observed values well (Fig. 4f). The impact of rainfall on SS variations decreased as the rainfall moved from the inlet to the transition zone and then to the lacustrine zone.

3.2. Effect of rainfall-induced turbidity currents on algal growth and light extinctions

3.2.1. Effect of turbidity on the algal biomass

We compared the simulation results for algal Chl-a with and without considering the influence of turbid water at three locations: riverine (DC6), transition (DC5), and lacustrine (DC2). Statistical analyses were performed on the time series results. At all three points, Chl-a increased by 44 %–58 % when turbid water was not considered compared to when it was considered. The difference in peak concentration was minimal (14 %) in the riverine zone but increased to 37 % and 57 % in the transition and lacustrine zones, respectively. Although the change in SS concentration was greatest in the riverine zone, leading to the expectation of the most significant Chl-a change there, the actual results showed larger Chl-a variations in the transition and lacustrine zones. This is likely due to algal growth during the longer retention time as the water moves downstream in the reservoir. The transitional and

lacustrine zones, which exhibit the physicochemical characteristics of a lake, showed greater changes in Chl-a (Fig. 5, Table 2).

The Riverine zone showed significant differences between May and July, the transition zone between May and October, and the lacustrine zone between September and October, appropriately reflecting the spatial changes from upstream to midstream and downstream. Suspended solids (SS) cause turbidity in water, scatter light entering the reservoir, increase light extinction, and inhibit algal growth. The model adequately reproduced this phenomenon.

Furthermore, the statistical analysis demonstrated that incorporating SS into the model significantly improved the prediction accuracy of Chl-a. Specifically, the error reduction rate (RMSE ERR) is 52–70 %, MAPE ERR by 52–67 %, when comparing results with and without SS consideration (Ghosh and Hall, 2008). These results illustrate the critical role of turbidity in regard to improving the reliability of algal bloom predictions (Table 3). Given that the change in the Chl-a concentration is significant when turbidity is not considered, accounting for the influence of turbid water on water quality modeling is necessary.

3.2.2. Light extinction variation by scenario

To further analyze the effect of turbid water on Chl-a, we analyzed

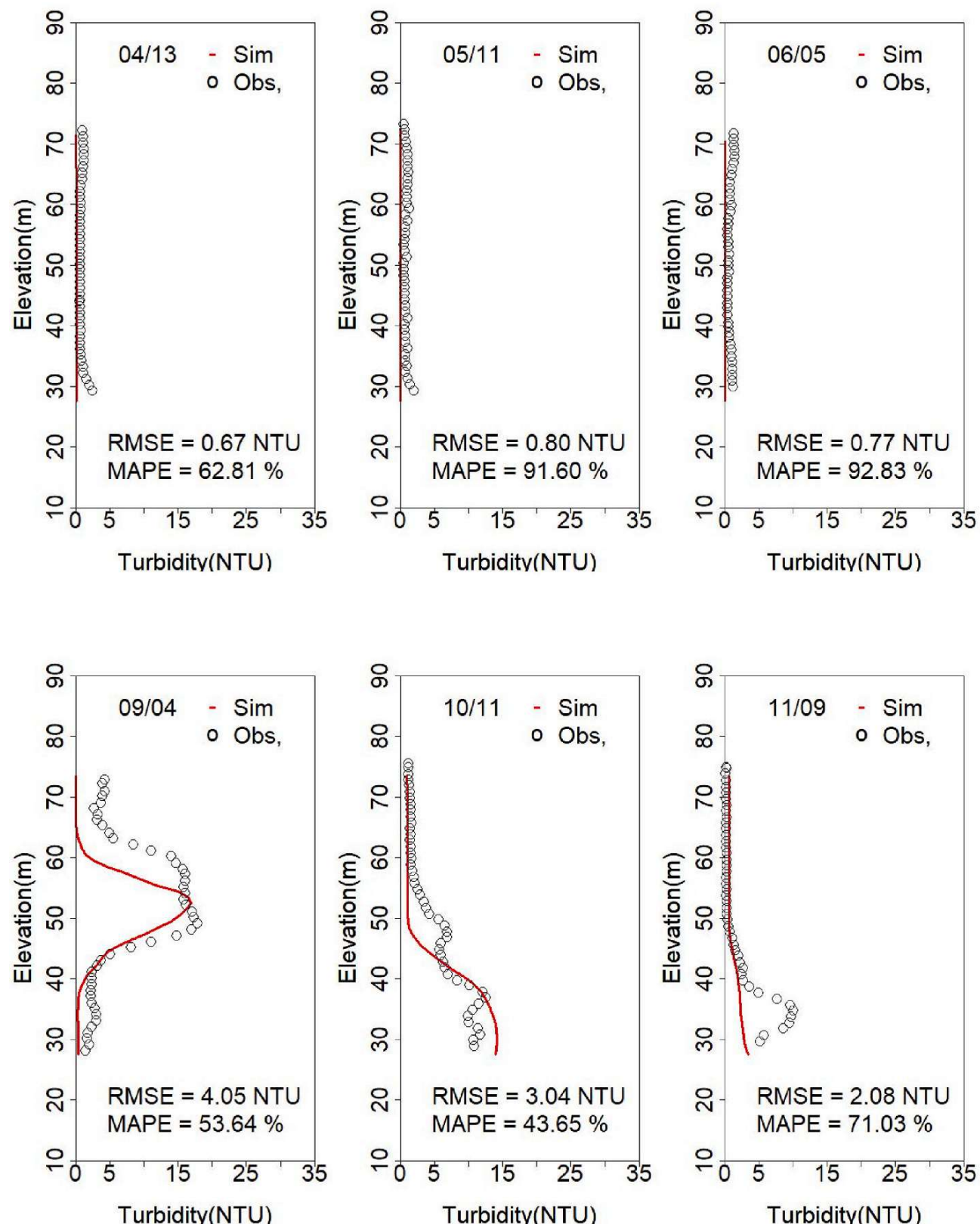


Fig. 3. Comparison of the observed and simulated turbidity profiles at the DC2 station.

the light extinction coefficient in terms of a time series at the surface, cross-section (curtain views), and sheet views. The light extinction coefficient was calculated based on the Chl-a, SS, DOC, and POC. Notably, the change in Chl-a due to the consideration of turbid water affected light extinction through the self-shading effect of algal communities. In the Riverine zone, during the periods of high Chl-a, the light extinction coefficient increased by a maximum of 4.28 m^{-1} (84 %) and an average of 64 % when including the spatiotemporal distribution of SS in the model. In the transition zone, it increased by a maximum of 3.93 m^{-1} (63 %) and an average of 81 %. The lacustrine zone showed the lowest difference in the light extinction coefficient among the three points, with a maximum increase of 1.84 m^{-1} (56 %) and an average of 50 % (Fig. 6, Table 4). While the difference in Chl-a was greatest in the lacustrine zone, depending on turbid water considerations, the light extinction coefficient showed the largest difference in the riverine zone. This is likely because the Chl-a concentration is the highest in the riverine zone, resulting in a greater shading effect, and it is the area directly affected by SS before settling occurs, which was calculated using Stokes' law in the model. As the turbid water moved along the middle layer density current downstream of the reservoir, there was little change in the light extinction coefficient owing to the SS at the surface of the lacustrine

zone. The curtain and sheet views show that the light extinction coefficient appears from the surface to a depth of approximately 5 m before the representative point of the transition zone (DC5), depending on the incoming SS. Following the heavy rainfall in July, high values were observed at the surface up to the riverine zone (DC6), followed by a mid-layer density current. Therefore, after July, no significant difference in the light extinction coefficient was observed at the surface due to the SS (Fig. 6, Figs. S6 and S7).

4. Discussion

Algal growth in water bodies is influenced by complex interactions between environmental factors including nutrients, light, and temperature. While much of the research on algal growth has been focused on nutrient-driven eutrophication, particularly the role of phosphorus as a limiting factor (Schindler, 1977; Klausmeier et al., 2004; Robson, 2014), less attention has been paid to the role of turbidity, especially in monsoon climates. Intense precipitation during the monsoons leads to increased turbidity through soil erosion, which can reduce light penetration and limit photosynthesis, thereby potentially restricting algal growth. While such a light limitation may have a minimal impact on

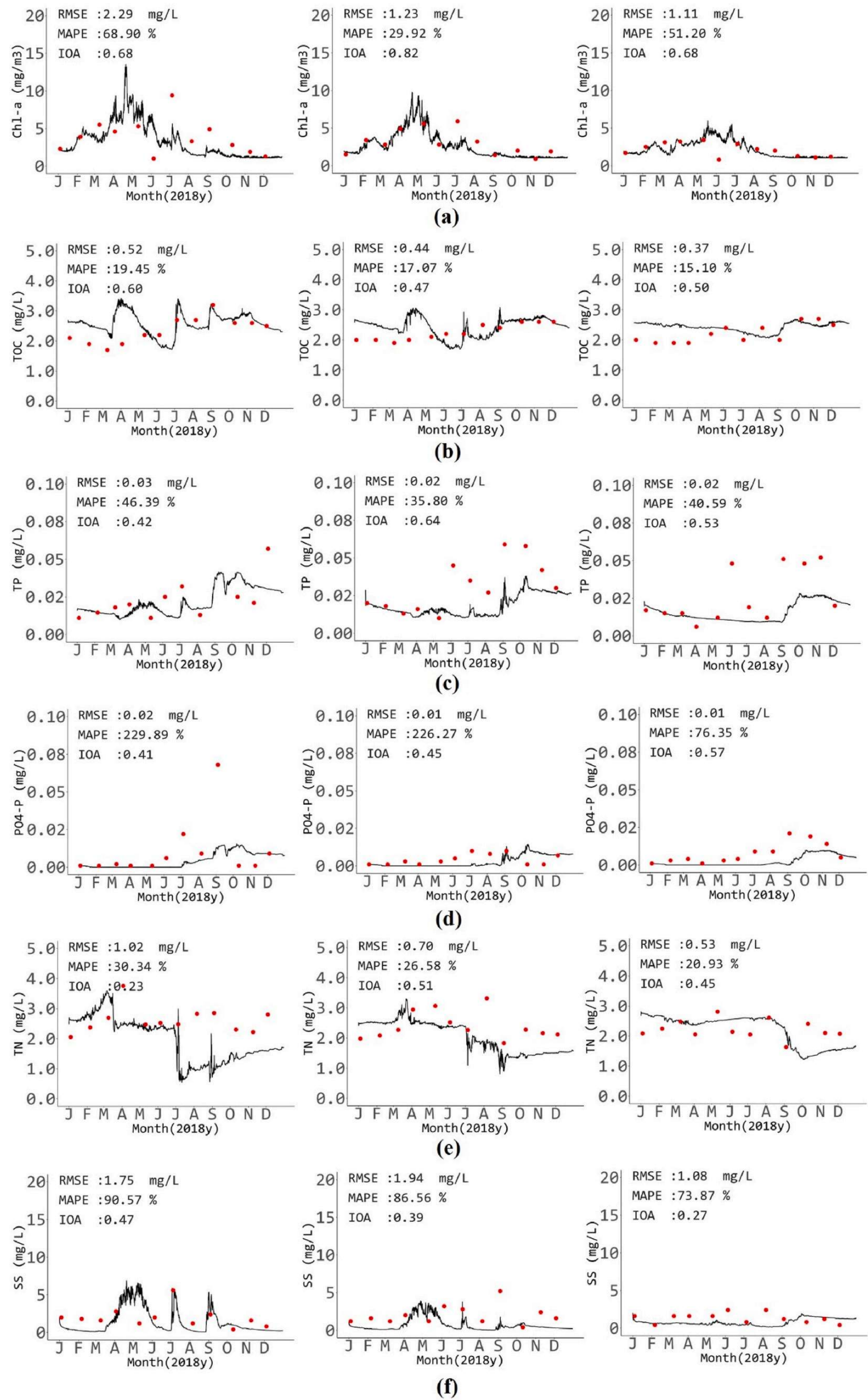
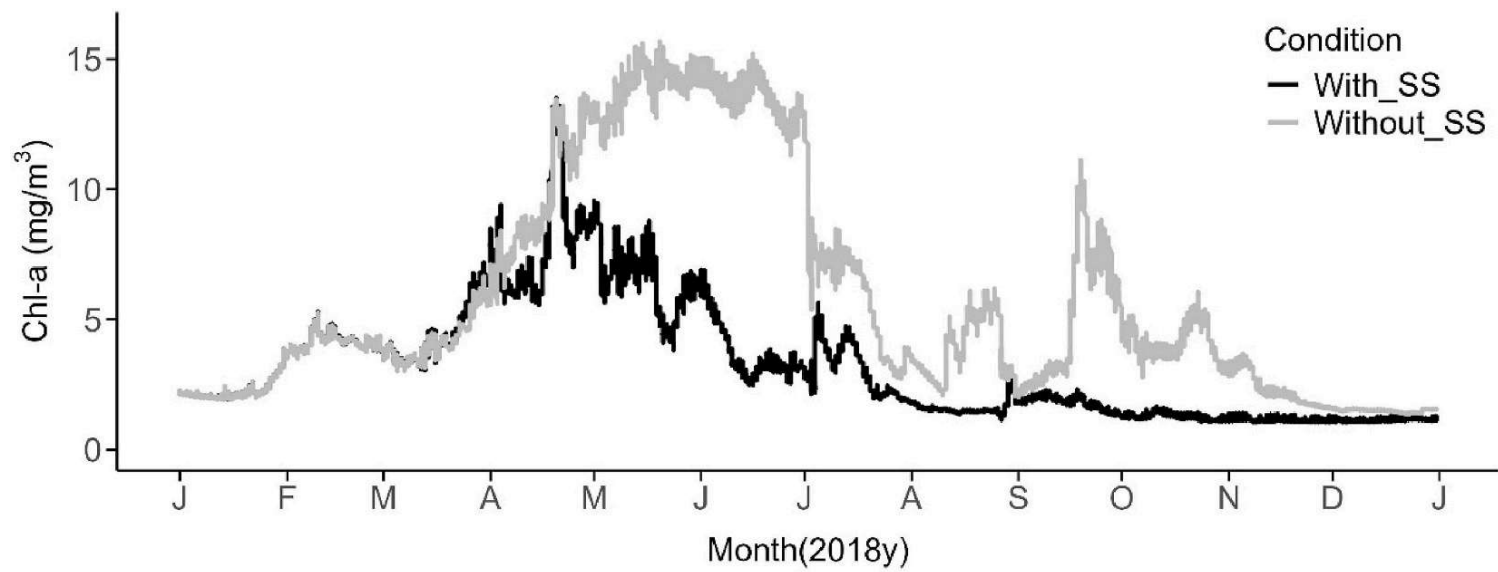
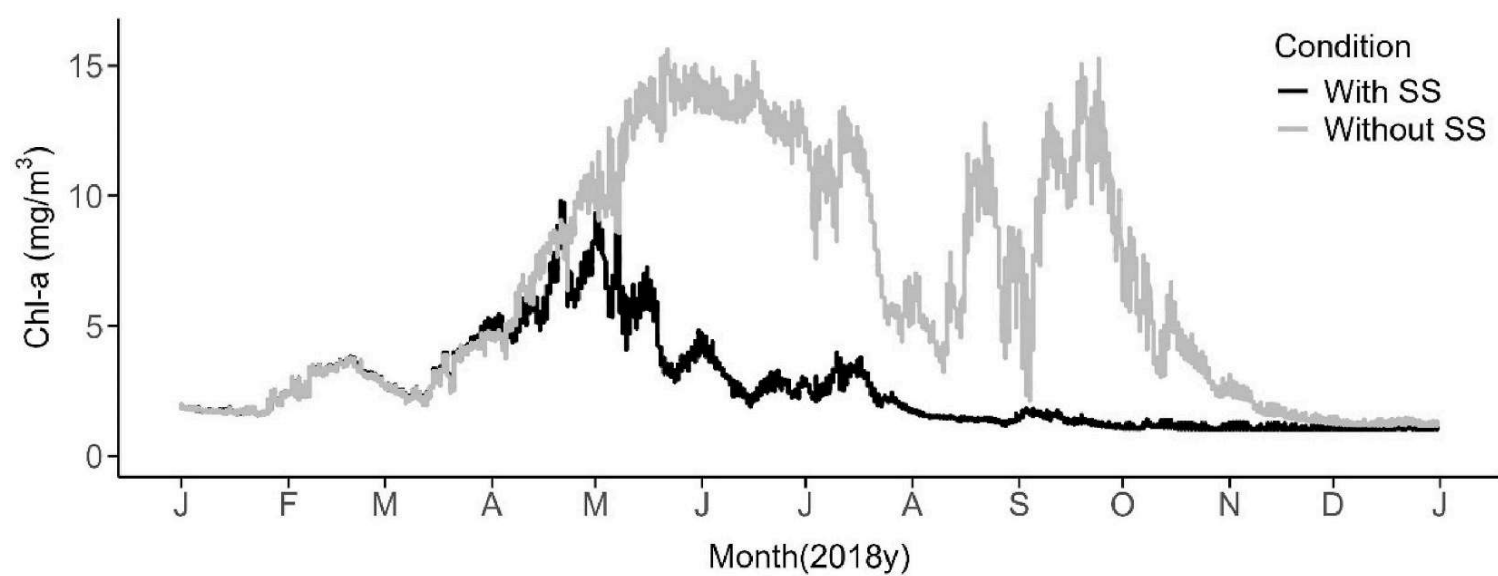


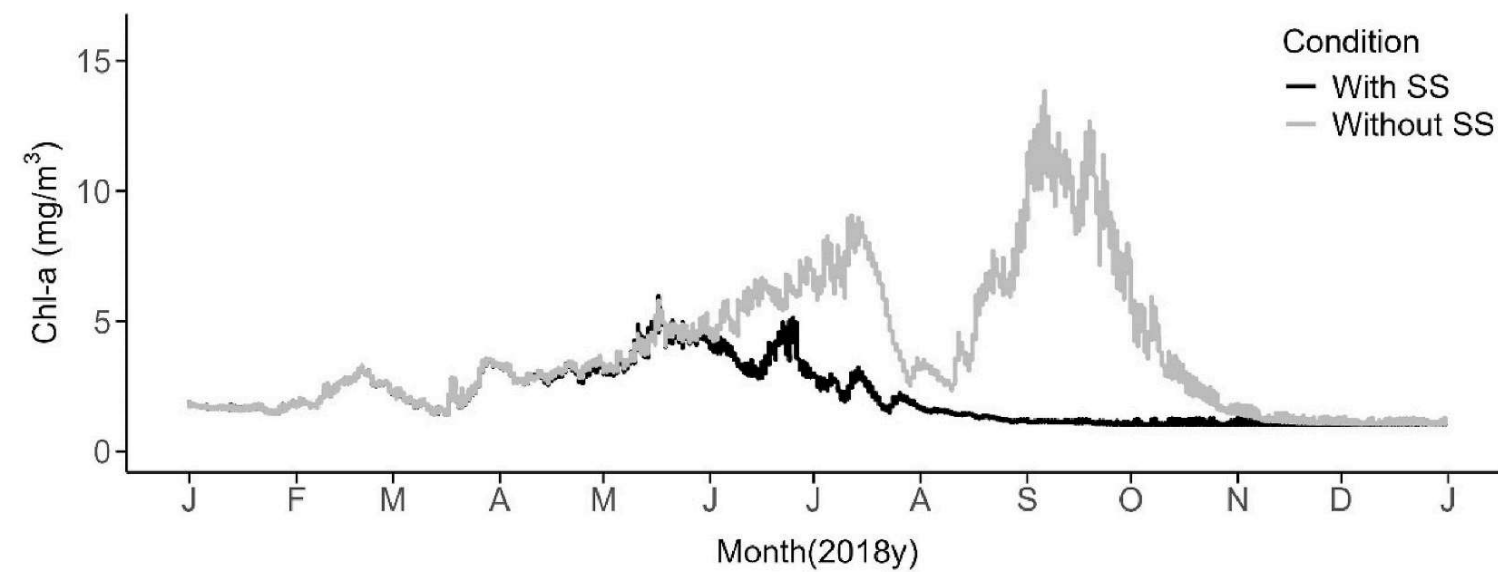
Fig. 4. Comparison of the simulated and observed water quality time series: (a) Chl-a, (b) TOC, (c) TP, (d) PO4-P, (e) TN, and (f) SS; riverine zone (left), transition zone (middle), and lacustrine zone (right).



(a) Riverine zone



(b) Transition zone



(c) Lacustrine zone

Fig. 5. Temporal changes in Chl-a concentrations in the (a) riverine zone, (b) transition zone, and (c) lacustrine zone of Daecheong Reservoir with and without considering the influence of turbid water.

rivers with short retention times, it can significantly affect the growth of algae in reservoirs with longer retention periods. In the present study, we have examined the effects of rainfall-induced turbidity on algal growth in a stratified reservoir. Specifically, we evaluated the spatio-temporal variability of light limitation and other growth-limiting factors within the reservoir and identified turbidity as a key factor influencing algal dynamics. Our light extinction findings were compared with those of previous studies, revealing differences that highlight the unique

challenges posed by turbidity in monsoon-affected reservoirs. These results underscore the importance of considering turbidity along with traditional nutrient management strategies in such environments. However, the limitations of our modeling approach, such as the lack of comprehensive measurement data, suggest that further research is required. Future studies should explore the temporal variability in turbidity and its interactions with other ecological variables to improve our understanding of algal growth dynamics in these complex systems.

Table 2
Descriptive statistics of the hourly Chl-a concentrations (mg/m³) simulated with and without accounting for SS, including the number of observations (Nobs), lower control limit (LCL), upper control limit (UCL), and standard deviation (Stdev).

Statistics	Riverine zone		Transition zone		Lacustrine zone	
	With SS	Without SS	With SS	Without SS	With SS	Without SS
Nobs	8760					
Minimum	1.05	1.35	1.02	1.15	1.02	1.03
Maximum	13.51	15.67	9.79	15.60	5.96	13.82
1.Quartile	1.41	2.66	1.22	2.33	1.13	1.70
3.Quartile	4.29	7.90	3.35	10.42	2.89	5.37
Mean	3.25	5.89	2.60	6.30	2.09	3.81
Median	2.23	4.09	1.98	4.76	1.71	3.03
LCL Mean	3.20	5.80	2.57	6.21	2.06	3.76
UCL Mean	3.30	5.98	2.64	6.39	2.11	3.87
Variance	5.27	18.42	3.08	19.87	1.16	6.98
Stdev	2.30	4.29	1.76	4.46	1.08	2.64

Table 3
Comparison of the prediction accuracy metrics (RMSE, MAPE, ERR) for simulated Chl-a (mg/m³) in different zones (Riverine, Transition, Lacustrine) with and without SS consideration.

$$ERR(\%) = \frac{(Error(WithoutSSmodel) - Error(WithSSmodel))}{Error(WithoutSSmodel)} \times 100$$

Statistics	Riverine zone		Transition zone		Lacustrine zone	
	With SS	Without SS	With SS	Without SS	With SS	Without SS
RMSE (mg/m ³)	2.29	4.72	1.23	4.15	1.11	3.34
MAPE (%)	68.90	142.62	29.92	90.68	51.2	123.47
RMSE-ERR* (%)	51.5		70.4		66.8	
MAPE-ERR* (%)	51.7		67.0		58.5	

4.1. Consideration of light extinction

The reproducibility of light extinction is crucial for accurately predicting the spatiotemporal distribution of algae in stratified reservoirs such as the Daecheong Reservoir. Various researchers have developed equations to calculate light extinction, primarily as a function of the Secchi depth (Williams et al., 1980; Brown, 1984; Chapra, 1997; Armengol et al., 2003; Caffrey et al., 2007). Although the Secchi depth shows a strong correlation with light extinction owing to its close association with the SS, existing equations have several limitations. First, the Secchi depth, as a simple indicator of water transparency, fails to fully reflect the complex light attenuation processes. Second, it represents only the average transparency in the visible light spectrum. Equations based solely on normal flow conditions may underestimate rapid changes during rainfall events. Third, although the Secchi depth indicates the total light attenuation by various substances in water, it is challenging to distinguish the individual contributions of different materials (SS, algae, and dissolved organic matter). Although SS significantly influences light attenuation, excessive algal growth and colony formation can increase light extinction. Fourth, the measurements may lack objectivity owing to variations in the observer’s eyesight, experience, and ambient light conditions. The Secchi disk relies on human visual judgment, which may underestimate the impact of fine particles. Consequently, in waterbodies with long retention times, it is crucial to consider the light-extinction effects of fine particles (both particulate and dissolved matter) (Bowers et al., 2020). Finally, there are limitations in quantifying light attenuation, particularly in cases of very high or low transparency where accurate measurements are difficult. Therefore, a

comprehensive consideration of various water quality parameters (such as SS, Chl-a, DOM, and POM) is essential for a more accurate estimation of light attenuation.

In this study, instead of using the light extinction coefficient calculated from the Secchi depth, we calculated the light extinction coefficient dynamically by considering the SS, Chl-a, DOC, and POC, which vary spatiotemporally within a water body. During the modeling process, we compared the dynamically calculated light extinction values in the riverine, transitional, and lacustrine zones with the relationships developed by previous researchers (Table 5), as shown in Fig. 7. All of the four relationship equations used only the Secchi depth in the first- or higher-order equations. At all points, the four model equations tended to underestimate light extinction compared with the AEM3D model results. Therefore, to improve the spatiotemporal simulation and prediction performance for algae, it is crucial to accurately predict the spatiotemporal distribution of various factors affecting light extinction, including turbidity.

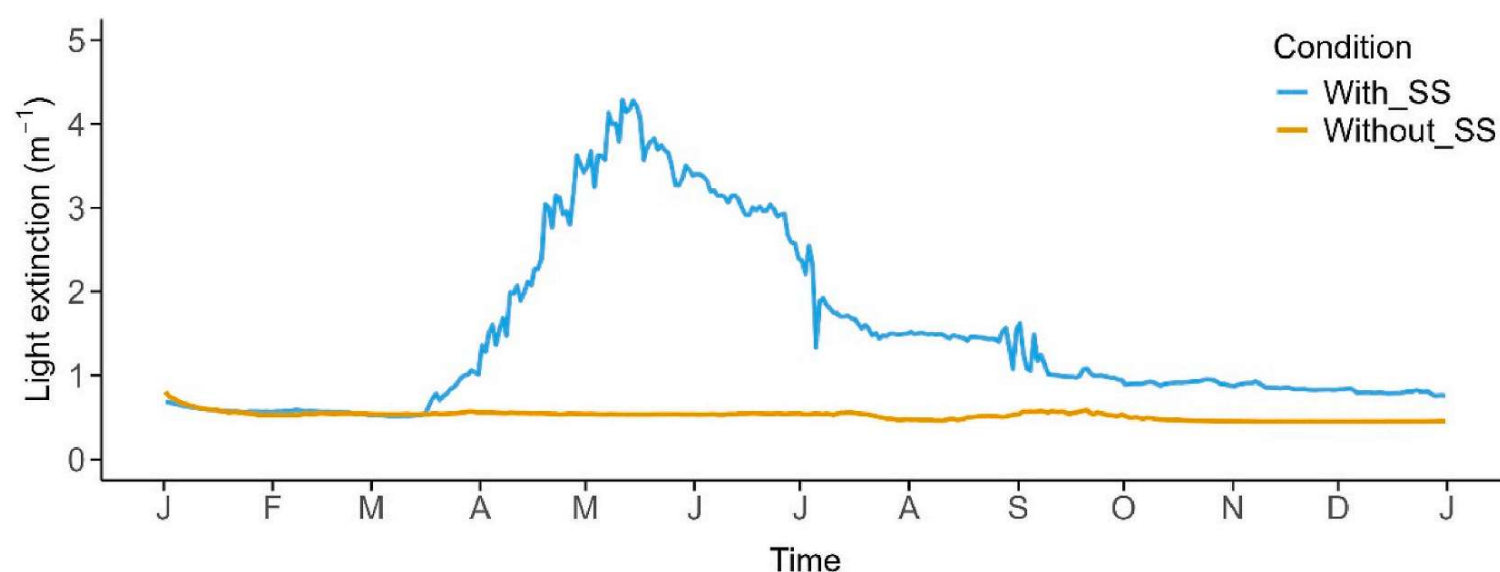
4.2. Temporal and spatial variations of limiting factors for different algal groups

The results presented above demonstrated that light exerts a significant influence as a limiting factor for algal growth, as evidenced by the comparison of Chl-a concentrations and light extinction with and without considering the effects of turbid water. However, algal growth is simultaneously constrained by additional factors, including nitrogen, phosphorus, and water temperature. Therefore, a quantitative assessment of the spatiotemporal impacts of each limiting factor is necessary. We present the spatiotemporal variability of growth-limiting factors (daily averaged light, nitrogen, phosphorus, and temperature) for cyanobacteria in Fig. 8 (diatoms and green algae are shown in Figs. S8 and S9). In these figures, the solid lines represent the limiting factor values when the effects of turbid water are considered, while the dashed lines show the results without such consideration.

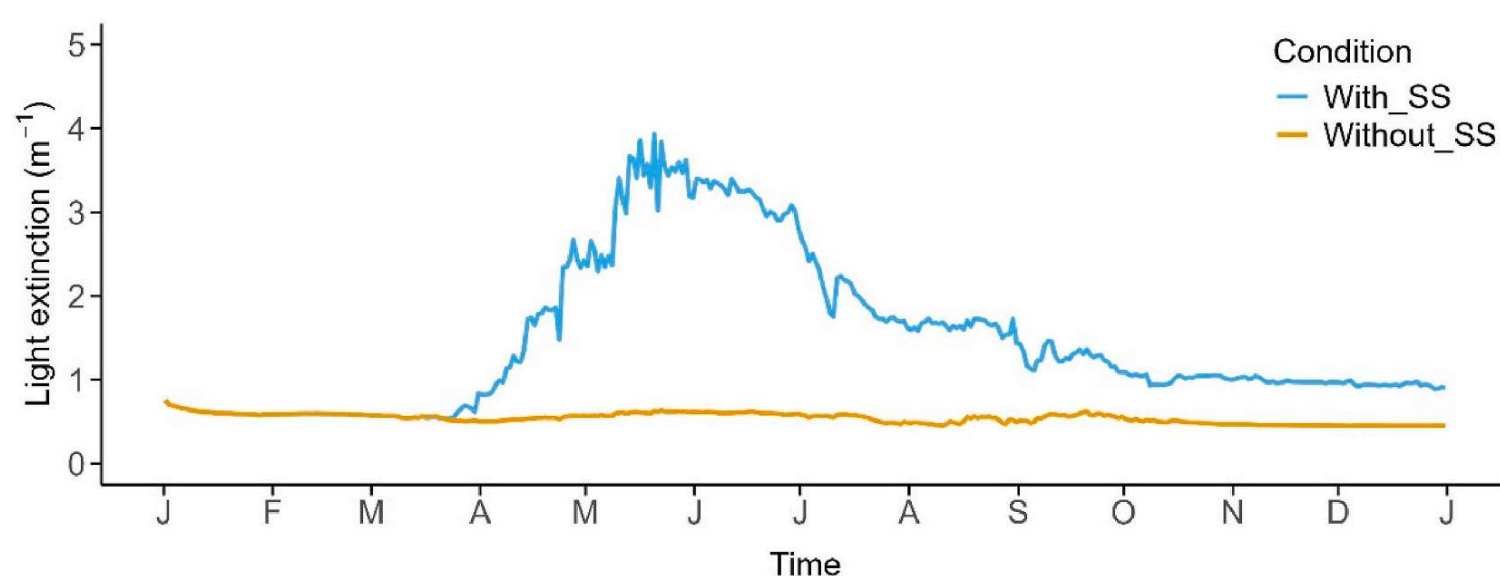
The three algal groups exhibited similar responses to light, nitrogen, and phosphorus limitations. In contrast, temperature limitation varied significantly among species due to their differing thermal preferences (Van der Grinten et al., 2005). Cyanobacteria, which favor high water temperatures, typically appear from June to October, whereas diatoms, which thrive in cooler conditions, emerge in April and drive spring blooms. At elevated temperatures, diatom growth rates are minimal, leading to their reduced presence when cyanobacteria dominate. Green algae emerge during the transition from spring to summer, subsequent to the appearance of diatoms.

When turbid water was not considered, light limitation was substantially reduced, allowing for increased algal growth and leading to significant phosphorus consumption until summer. Under these conditions, phosphorus acted as a significant limiting factor, as suggested by low limiting factor values. However, after July, phosphorus limitation appeared to weaken due to its influx during rainfall events. In contrast, when the effects of turbid water were incorporated into the model, light limitation became more pronounced, particularly in the riverine zones, as the inflowing turbid water significantly reduced light availability. Downstream, as the inflowing turbid water settled along the middle-density current, its influence on surface layer growth limitation appeared to diminish.

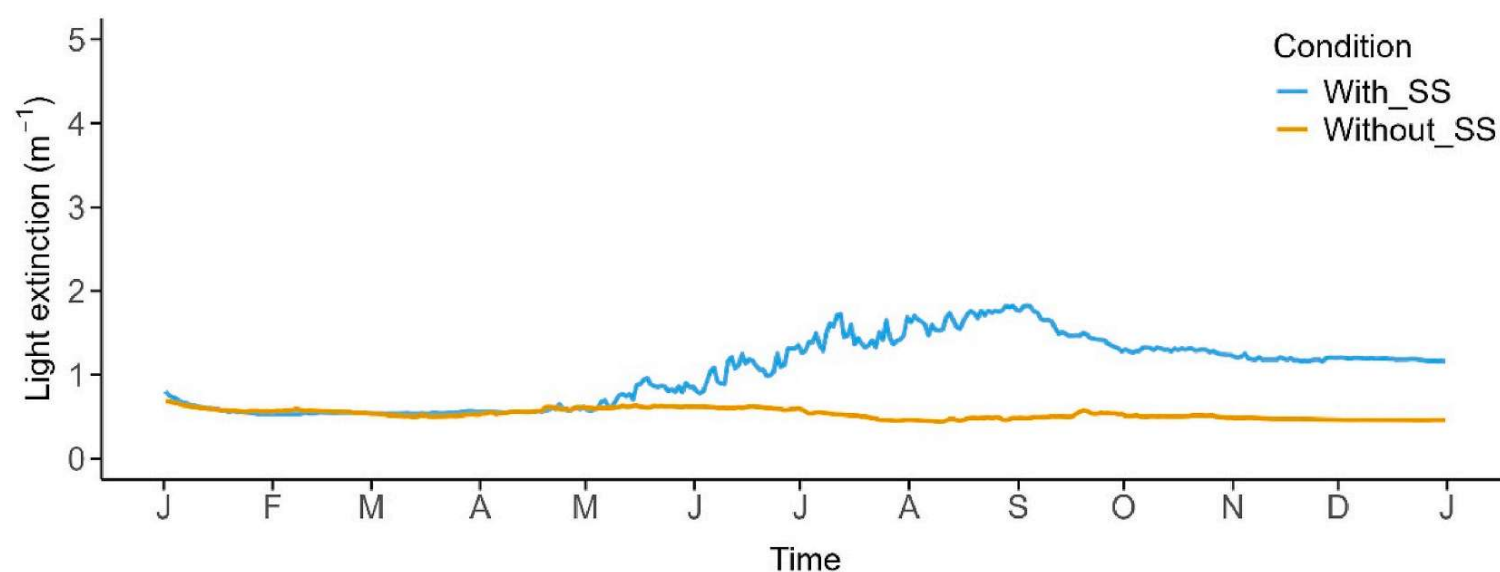
Model results indicated that phosphorus was more likely to act as a limiting factor than nitrogen throughout the year, with phosphorus briefly limiting growth during the spring bloom before its influence waned. To further address nutrient limitation, we compared the time series of measured dissolved inorganic nitrogen (NO₃-N and NH₄-N) and dissolved inorganic phosphorus (PO₄-P) at three monitoring sites in Daecheong Reservoir with their respective half-saturation constants (Fig. S10). The analysis revealed that until May, DIP concentrations were near the half-saturation threshold (Table S5), suggesting significant phosphorus limitation. However, subsequent rainfall and runoff



(a) Riverine zone



(b) Transition zone



(c) Lacustrine zone

Fig. 6. Comparison of light extinction in the (a) riverine zone, (b) transition zone, and (c) lacustrine zone of Daecheong Reservoir with and without considering the influence of turbid water.

events introduced non-point source inputs that elevated DIP concentrations well above the half-saturation constant, thereby alleviating phosphorus limitation during the summer. In contrast, although nitrogen concentrations exhibited transient declines during rainfall events in March and July–August, they did not fall to levels indicative of limitation in the model simulations. Furthermore, the box plots in Fig. S11, which illustrate a 10-year (2015–2024) dataset of measured Chl-a, DIP, and DIN across the riverine, transition, and lacustrine zones, confirm

that DIN consistently exceeded the nitrogen half-saturation constant (0.02 mg/L) (United States Environmental Protection Agency [USEPA], 1985; Cole and Wells, 2006; Hipsey et al., 2006). This suggests that nitrogen concentrations were generally sufficient to prevent limitation throughout the year.

While these findings are derived primarily from model outputs and are thus constrained by the limited observational data available for validation, recent investigations in monsoon-climate reservoirs that

Table 4
Descriptive statistics of 12-hourly light extinction (m^{-1}) simulated with and without accounting for SS, including the number of observations (Nobs), lower control limit (LCL), upper control limit (UCL), and standard deviation (Stdev).

Statistics	Riverine zone		Transition zone		Lacustrine zone	
	With SS	Without SS	With SS	Without SS	With SS	Without SS
Nobs	720					
Minimum	0.51	0.44	0.54	0.45	0.52	0.45
Maximum	4.28	0.68	3.93	0.76	1.84	0.80
1.Quartile	0.79	0.48	0.83	0.49	0.57	0.47
3.Quartile	1.77	0.58	1.80	0.59	1.32	0.55
Mean	1.48	0.53	1.48	0.54	1.04	0.52
Median	0.98	0.52	1.06	0.55	1.16	0.54
LCL Mean	1.41	0.53	1.41	0.54	1.01	0.52
UCL Mean	1.55	0.54	1.54	0.55	1.07	0.52
Variance	1.06	0.00	0.83	0.00	0.17	0.00
Stdev	1.03	0.06	0.91	0.06	0.41	0.05

Table 5
Equations for estimating light extinction based on Secchi depth, as proposed in previous studies.

Researcher	Extinction coefficient (λ)	Water body
Chapra	$\lambda = \frac{\alpha}{Z_{\text{secchi}}}$ ($\alpha = 1.4\text{--}1.9$, typical value 1.7)	Fresh water
Williams	$\lambda = \frac{1.11}{Z_{\text{secchi}}^{0.73}}$	Lakes and reservoirs in Ohio, USA
Armengol	$\lambda = \frac{1.36}{Z_{\text{secchi}}^{0.86}}$	Sau Reservoir, Spain
Caffrey	$\text{Log } \lambda = -0.96 \log (Z_{\text{secchi}} + 0.3)$	32 lakes in Florida, USA

highlight the critical role of light limitation have independently reached conclusions that align closely with our own. For instance, a recent investigation of 54 lakes and reservoirs in the eastern plains of China reported that only 16 % were nutrient-limited, whereas 84 % were constrained by non-nutrient factors (Janatian et al., 2020; Zou et al., 2022). In reservoirs characterized by extensive catchment areas and prolonged retention times, algal growth appears only marginally restricted by nitrogen and phosphorus, whereas turbidity-induced light limitation during rainfall events exerts a more pronounced influence. Consequently, failing to account for the effects of turbid inflows or resorting to excessive parameter calibration in conventional water quality models can introduce considerable uncertainties. To achieve robust simulations of algal dynamics and water quality, it is therefore crucial to incorporate accurate representations of turbid water effects, appropriately tailored to the specific characteristics of each water body.

Our model-based evaluation provides useful insights into the

interplay between light and nutrient limitation, but it remains largely theoretical. The inability of the current approach to distinctly separate nitrogen and phosphorus limitation, or to account for the common occurrence of co-limitation is primarily due to limitations in the measured data and the challenges in fully validating the DIP and DIN components of the model. Therefore, further phytoplankton bioassays, coupled with more rigorous model calibration and validation, are essential for definitively assessing the nutrient limitations of algal growth across different temporal scales.

4.3. Implications of the research findings for HAB management in dam reservoirs

The findings of this study have considerable implications for the management of HABs in dam reservoirs, particularly in the temperate monsoon climatic region of South Korea. By addressing the critical gaps in existing management strategies, this study provides a more advanced and comprehensive approach to water quality management. Below, we discuss the potential applications of these findings to practical management and policy development.

First, the development of a 3D modeling technique that incorporates the influence of turbid inflows on algal growth is a major advancement in water quality prediction. This model enables reservoir managers to accurately assess the impact of rainfall-induced turbidity on algal bloom dynamics. By integrating rainfall forecasts into this model, managers can more effectively predict algal bloom patterns and develop proactive strategies to mitigate HABs. This predictive approach represents a significant improvement over traditional reactive management, potentially reducing the economic and ecological impacts of HABs.

Moreover, the ability of the model to analyze the spatiotemporal variability of algal growth-limiting factors such as light, temperature, and nutrients offers a deeper understanding of HAB dynamics. This multifactor analysis allowed managers to identify the primary drivers of algal blooms under varying conditions. For example, if light limitation emerges as a critical factor during certain periods, artificial shading can be used to control algal growth. Conversely, if nutrient availability is the dominant factor, management efforts should focus on reducing nutrient inputs through targeted watershed management practices. This precision in identifying and addressing the key factors driving HABs enables more efficient resource allocation and enhances the effectiveness of control measures.

In addition, this study challenges the prevailing focus on reducing the TP as the primary strategy for HABs management in South Korean reservoirs. Although nutrient management is essential, our findings highlight the need for a holistic approach. This model allows policy-makers to prioritize and diversify their management strategies by quantifying the impact of various factors on algal blooms. This could

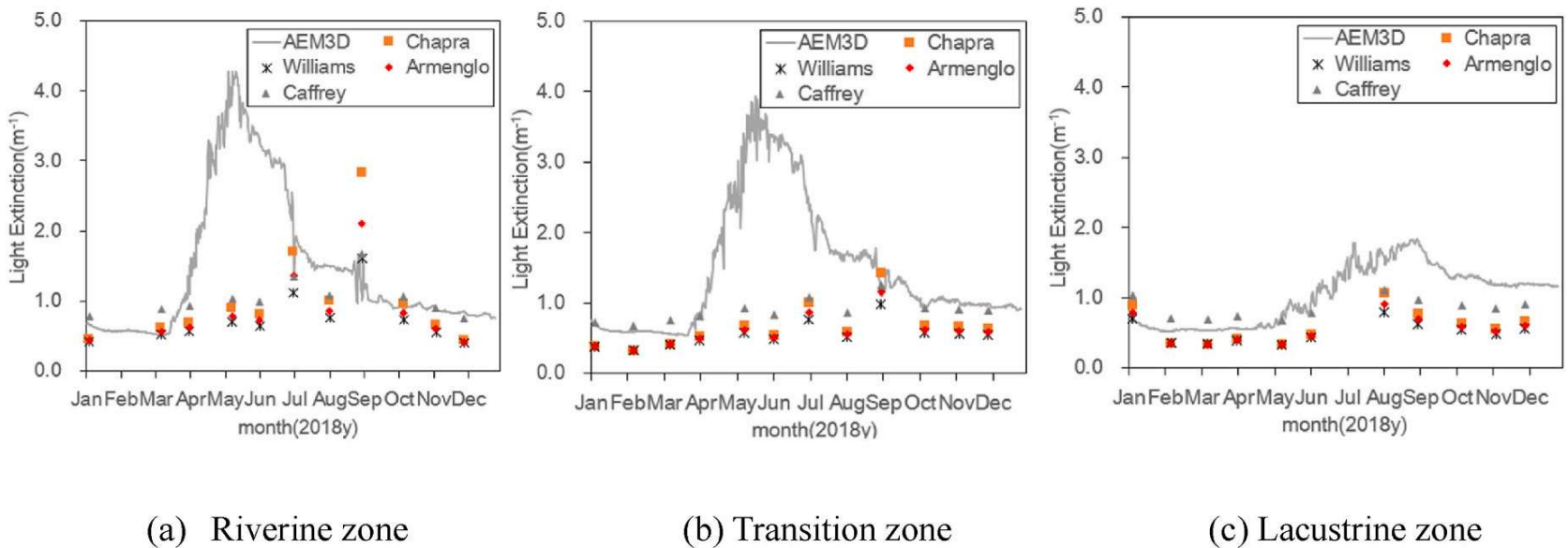


Fig. 7. Comparison of AEM3D light extinction coefficient with other models based on Secchi depth measurements in (a) riverine zone, (b) transition zone, and (c) lacustrine zone.

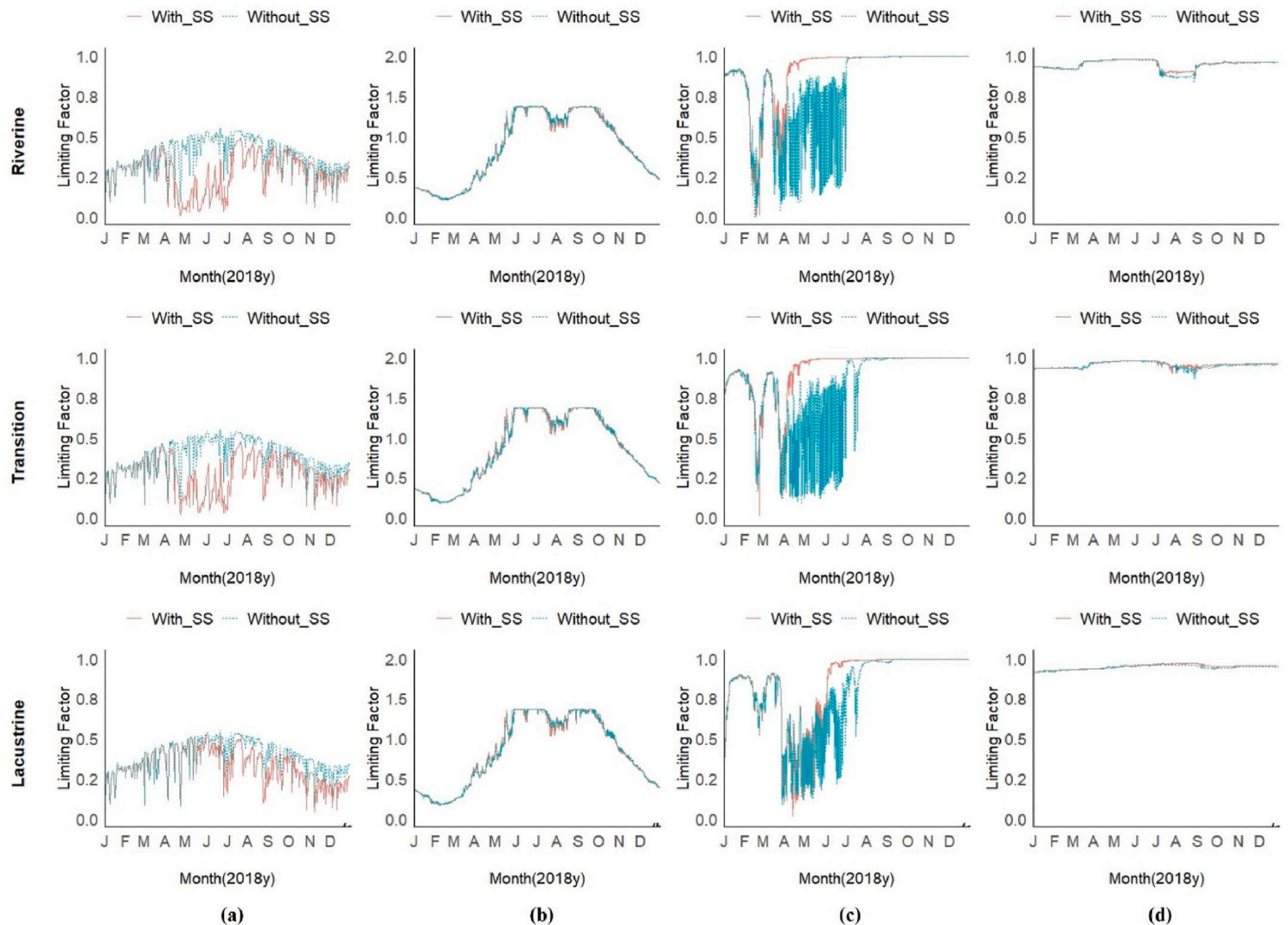


Fig. 8. Temporal variation in limiting factors in the riverine (top), transition (middle), and lacustrine (bottom) zones for cyanobacteria: (a) light (daily average), (b) temperature, (c) phosphorus, and (d) nitrogen.

lead to the development of seasonally and regionally tailored HABs management plans, optimizing the use of limited resources and improving overall management outcomes.

Finally, this study provides valuable insights into long-term adaptations to climate change. By incorporating rainfall patterns and their effects on turbidity and algal growth into the model, future water quality can be projected under different climate change scenarios. This capability will enable policymakers to develop forward-looking water quality management policies and prepare adaptive strategies for potential climate-induced changes in water quality.

4.4. Limitations of research and future improvement

This study emphasizes the importance of light limitations in predicting HABs and the necessity of considering the effects of turbid water on the study reservoir. However, an important limitation of this study is the lack of direct validation of the light extinction coefficients (K_e) used in the model. Although the AEM3D model employed in this study incorporated the spectral absorption and scattering coefficients for various water constituents (e.g., DOC, POC, aggregates, and suspended solids) at different wavelengths, the assigned K_e values were not validated against field measurements. This omission may influence the precision of the light-climate predictions of the model, particularly when capturing the nuanced effects of wavelength-specific light attenuation. Moreover, the average transparency of the visible-light spectrum used in the model does not fully reflect the wavelength-specific

interactions between light and water constituents, which is critical for a comprehensive understanding of light dynamics in aquatic systems. This simplified approach may limit the applicability of our findings to more complex and highly variable environments. Future studies should consider adopting advanced methodologies to overcome these limitations. Radiative transfer models and narrowband approaches can provide detailed insights into the spectral characteristics of light attenuation (Jarabo et al., 2018; Li et al., 2023). Additionally, the integration of hyperspectral remote-sensing data and also in situ spectral measurements could enable better parameterization and validation of wavelength-specific absorption and scattering coefficients. These advancements would allow models to more accurately represent the spatial and temporal variability of light conditions in reservoirs and other aquatic ecosystems. By incorporating these refined techniques, future studies could enhance our understanding of the complex interplay between light attenuation and water quality parameters, thereby offering more robust tools for the management of aquatic environments.

Water quality modeling involves various sources of uncertainty, primarily related to bathymetry data, boundary condition data, and model parameters. Regarding bathymetry data, a comparison with the measured reservoir capacity curve showed that the topography of the model adequately reflected the actual topographical characteristics. For the boundary condition data, we incorporated the daily measured flow rates and high-frequency water temperature input data generated using machine learning techniques into the model. However, for water quality data, we used monthly and weekly interval data, which have limitations,

especially when considering the high concentrations of organic matter and phosphorus flowing from the watershed during rainfall events. The tendency of the model to overestimate the SS concentrations at the dam inlet during rainfall events could be attributed to the scarcity of observational data during periods of elevated SS levels. This discrepancy arises from the low monitoring frequency at the inlet during precipitation events. Such temporal limitations in data collection pose significant challenges for the accurate observation and quantification of short-term fluctuations in SS concentrations associated with rainfall. To compensate for this, we applied changes in water quality concentrations during the dry and rainy periods, as suggested in the literature. Nevertheless, this remains a major factor that contributes to errors in model reproducibility. One approach for improving model reproducibility is to use high-frequency input boundary data. Continuous monitoring of the water quality parameters during rainfall events and dry weather conditions should be conducted using high-frequency sensors. The data collected in this manner can not only be used as model inputs but also as training data for machine learning models. This can potentially be utilized to generate water quality boundary data that reflect inflow characteristics during rainfall periods.

In addition to data-related challenges, unmeasured parameters pose another significant source of uncertainty in water quality models. Here, key parameters were largely drawn from published literature. Notably, for the algae, which are grouped into diatoms, green algae, and cyanobacteria, we relied on literature-based estimates of their proportions and corresponding parameters at the inflow boundary. While this approach is practical given limited observational data, it inevitably introduces uncertainty into the simulations. Enhancing model reliability therefore necessitates increased frequency of algal monitoring and more comprehensive parameter measurements. Such efforts would refine both the parameterization of algal growth and the representation of key processes—ultimately improving our ability to predict and manage HABs in dam reservoirs.

An important area for future research is the potential impact of catchment management strategies that reduce the loads of suspended sediments entering the reservoir. On one hand, a decrease in suspended sediments would improve water clarity, which could enhance light availability and, in turn, stimulate phytoplankton growth—particularly that of cyanobacteria. On the other hand, suspended sediments often carry particulate nutrients that, through mineralization and desorption processes, contribute to the dissolved nutrient pool. Consequently, reducing sediment loads may also lead to lower inputs of dissolved nutrients. The net effect on phytoplankton biomass, therefore, remains uncertain. Future studies should integrate catchment management scenarios into the modeling framework to assess whether the benefits of improved water clarity outweigh the potential drawbacks of diminished nutrient inputs, or vice versa. This integrative approach will yield a more comprehensive understanding of how alterations in sediment dynamics influence phytoplankton, and ultimately, reservoir water quality.

5. Conclusions

This study highlighted turbidity as a dominant factor controlling algal growth by limiting light penetration over extended periods in monsoon-affected reservoirs. In large reservoirs, such as the Daecheong Reservoir, where turbid water persists, light limitation due to turbidity outweighs nutrient limitation as the primary constraint on algal growth. This phenomenon results in a 44–58 % reduction in chlorophyll-a (Chl-a) concentrations from riverine to lacustrine zones. Furthermore, the integration of turbidity currents with nutrient dynamics in the modeling framework significantly enhanced HAB prediction accuracy, improving the RMSE by 51–70 % and the MAPE by 52–67 % compared to conventional nutrient-only models.

In this study, wavelength-specific light attenuation characteristics were considered by applying light attenuation coefficients to different

water quality parameters. However, a limitation of this study is its inability to compare the modeled light attenuation values with actual measured data, which is proposed as a direction for future research.

This study has provided a robust scientific foundation for the transition from traditional phosphorus-centric water management strategies to more holistic and adaptive approaches. By elucidating the interplay between turbidity, nutrients, and algal growth, this work will equip water managers and policymakers with the knowledge to implement more effective and sustainable HAB mitigation strategies in dam reservoirs in the monsoon region. Such strategies are becoming increasingly critical under the pressures of climate change and evolution of watershed conditions.

CRedit authorship contribution statement

Dongmin Kim: Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Sewoong Chung:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of generative AI and AI-assisted technologies in the writing process

The authors used Claude, an AI language model developed by Anthropic, to improve language and readability during the preparation for this study. After using this tool, the authors thoroughly reviewed and edited the content as required and took full responsibility for the content.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix B. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jenvman.2025.125291>.

Data availability

Data will be made available on request.

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